

COMPUTER ORGANISATION AND ARCHITECTURE ASSIGNMENT (12 QUESTIONS):

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Submission Date: 06 Aug, 2026

Q1. What is Yotta?

ANS: Yotta is a company that offers services for cloud computing and services for data center. Yotta features include:

- * Services of cloud computing
- * Security of data stored
- * Less use of energy in data centers
- * Options for recovery at time of disaster

Yotta is used for things like Artificial Intelligence, medical care, banking services, public services, education and research.

Q2. AI Summit: Ten Takeaways

1. Artificial Intelligence is being used in industries like healthcare, finance, education and manufacturing.
2. Artificial Intelligence can do tasks like creating content, designing graphics and helping with learning activities.
3. We have to make sure Artificial Intelligence systems are developed and used in a way so they are safe and fair.
4. Good quality data helps Artificial Intelligence models work better and be more accurate.
5. If you want to work with Artificial Intelligence you should learn programming and mathematics
6. Graphics Processing Units are important for training Artificial Intelligence

models.

7. Cloud computing lets us use computers without having to buy expensive hardware.

8. There are open-source Artificial Intelligence tools that are easy to use and free.

9. Artificial Intelligence is helping people in jobs and making work easier. 10. Since

Artificial Intelligence is always changing we have to keep learning things. **Q3.**

What is NPTEL?

ANS: NPTEL or National Programme on Technology Enhanced Learning is a learning platform supported by the Ministry of Education Government of India. You can watch video lectures from professors at IITs and IISc learn at your pace and get certificates when you finish a course. These certificates can help you get a job or an internship. You can learn from anywhere because the courses are online.

Q4. CPU vs GPU vs TPU

ANS: The CPU is used for computer work like running apps and the operating system. The GPU has cores and is good for graphics and doing many calculations at the same time. The TPU is a chip made for Artificial Intelligence work. The CPU is flexible the GPU is fast for calculations and the TPU is fast for Artificial Intelligence work.

Q5. Who is the founder of Intel and AMD?

ANS: Intel was founded on July 18 1968 by Robert Noyce and Gordon Moore. The headquarters is in Santa Clara, California, USA. AMD or Advanced Micro Devices, Inc was founded on May 1 1969 by Jerry Sanders and seven other people. The headquarters is also in Santa Clara, California, USA.

Q6. What is the Turing Award? Who received it recently?

ANS: The A.M. Turing Award is an important award in computer science. It is like the Nobel Prize of Computing because it recognizes people who have made a contribution to computing. Recently Andrew G. Barto and Richard S. Sutton received the award for their pioneering work.

Q7. Why is GPU faster than CPU?

ANS: The CPU is the brain of the computer and handles many tasks. The GPU is designed to do calculations at the same time especially for graphics and big data. That is why the GPU is faster than the CPU for some tasks.

Q8. What are the parameters to decide which programming language should be used to develop an application?

ANS: To choose a programming language you need to think about a

things:

- * What kind of application you are making

- * How. Secure it needs to be

- * What platform you are using

- * How easy it is to develop

For example Python is often used for Artificial Intelligence Java is used for Android apps. C++ is used for games and system software. Choosing the language helps your application work better and saves time.

Q9. The Imitation Game

ANS: After watching The Imitation Game I learned that being smart is not enough. You also need to be determined and patient. The movie shows Alan Turing using his knowledge and not giving up even when people did not believe in him. His work changed the world. I learned that working in a team is important. Turing was very smart. He could not have done it alone. It taught me that working together is better than working

Q10. What is the Stampeding Herd Problem?

The Stampeding Herd Problem happens when many threads or processes are waiting for one shared resource. When the resource becomes available they all try to access it at the time. This causes an overload on the CPU and slows down the system. It is like when many students are waiting for a classroom to open and when it does they all try to go at the same time. It gets crowded and slow.

Q11. What is the CPU organization of the processor used in our laptop?

ANS: The CPU organization is called Apple M4. It has a design with 10 cores: 4

high-performance cores and 6 efficiency cores. The high-performance cores are for tasks and the efficiency cores are for lighter tasks. This design gives both speed and good battery life.

Q12. What is Raspberry Pi?

ANS: Raspberry Pi is a computer, about the size of a credit card. It has a functional RAM, CPU, GPU and input/output devices. It can run operating systems like Raspberry Pi OS. Because it is very affordable it is used to make things, like honeypots, robots, networking tools and IoT devices.